

STAT 4005 - Time Series

Chapter 5 - Examples in R

Agenda

- 1 A general procedure for analyzing time series
- 2 Example 1 - Yield of government securities
- 3 Example 2 - Accidental death

A general procedure for analyzing time series

General procedure:

- 1 Draw a time series **plot** and ACF plot to check for stationarity
- 2 **Filtering/differencing** to remove trend and seasonal effect
- 3 Draw ACF/PACF **plot** to get some idea about **which ARMA models** to try
- 4 **Fit** several ARMA models and get the best one according to some order selection criteria.
- 5 Model **diagnostics**: time series/ACF/PACF plots for *residuals*; Ljung-Box test.
- 6 The time series is described by

Trend+Seasonal effect+Selected ARMA model

Example 1 - Yield of government securities

This is an example of the yield of short-term government securities for 21 years for a country in Europe in a period in the 1950s and 1960s. The data are stored in the file `yields.dat`.

- Data: $\{Y_t\}$
- R-code:

```
yield.ts=scan("yields.dat")  
par(mfrow=c(2,2))  
ts.plot(yield.ts)  
acf(yield.ts)  
pacf(yield.ts)
```

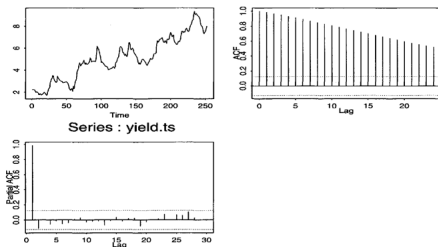


Fig. 5.1 Time series and ACF for the yield data.

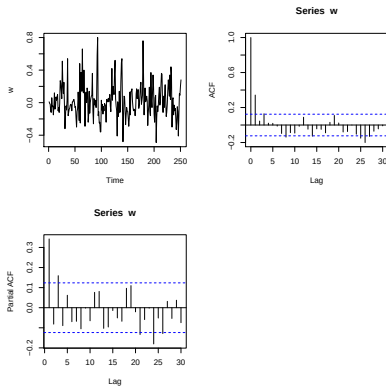
data can be download from
<http://www.sta.cuhk.edu.hk/NHCHAN/TSBook2nd/dataset.html>

Example 1 - Removing Trend

- To remove the upward trend in the data, we do differencing: $W_t = (1 - B)Y_t$

R-code:

```
w=diff(yield.ts)
par(mfrow=c(2,2))
ts.plot(w)
acf(w,30)
pacf(w,30)
```



- ACF and PACF suggest ARIMA(0,1,1)

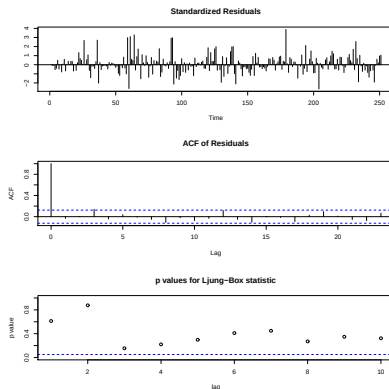
Example 1 - Model Fitting and Diagnostic checks

ARIMA(0,1,1) Model: $W_t = (1 - B) Y_t = Z_t + \theta Z_{t-1}$

R-code:

```
w.1=arima(w,order=c(0,0,1),  
method="ML", include.mean=F)  
w.1$coef  
ma1  
0.4349813  
tsdiag(w.1)
```

Result: $W_t = Z_t + 0.43Z_{t-1}$



Example 1 - Model Fitting and Diagnostic checks

ARIMA(0,1,1) Model: $W_t = (1 - B)Y_t = Z_t + \theta Z_{t-1}$

Alternatively:

R-code:

```
w.2=arima(yield.ts,  
order=c(0,1,1),method="ML")
```

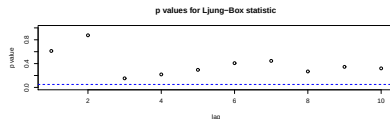
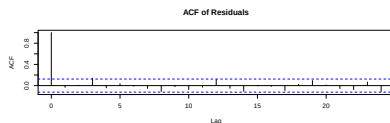
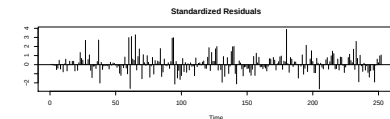
```
w.2$coef
```

```
ma1
```

```
0.434981
```

```
tsdiag(w.2)
```

Result: $W_t = Z_t + 0.43Z_{t-1}$



Example 1 - Prediction

Model:

$$\text{ARIMA}(0,1,1): W_t = Y_t - Y_{t-1} = Z_t + 0.43Z_{t-1}$$

Prediction:

$$\begin{aligned}\hat{Y}_{n+1} &= Y_n + \hat{Z}_{n+1} + 0.43\hat{Z}_n \\ &= Y_n + 0.43\hat{Z}_n \quad (\text{predict } \hat{Z}_{n+1} \text{ by } 0)\end{aligned}$$

$\hat{Z}_n$ can be obtained by iterating $Z_t = Y_t - Y_{t-1} - 0.43Z_{t-1}$

- $\hat{Z}_1 = Y_1 - Y_0 - 0.43\hat{Z}_0 = Y_1$
- $\hat{Z}_2 = Y_2 - Y_1 - 0.43\hat{Z}_1$
-
- $\hat{Z}_n = Y_n - Y_{n-1} - 0.43\hat{Z}_{n-1}$

Example 1 - R-code to obtain $\hat{Z}_n$

- ARIMA(0,1,1): $W_t = Y_t - Y_{t-1} = Z_t + 0.43Z_{t-1}$
- $\hat{Z}_n$ can be obtained by iterating $Z_t = Y_t - Y_{t-1} - 0.43Z_{t-1}$

R-code to obtain $\hat{Z}_n$

```
yield.ts=scan("yields.dat")
n=length(yield.ts)
z.hat=rep(0,n)
z.hat[1]=yield.ts[1]
for(i in 2:n){
  z.hat[i]=yield.ts[i]-yield.ts[i-1]-as.numeric(w.2$coef[1])*z.hat[i-1] }
prediction=yield.ts[n]+as.numeric(w.2$coef[1])*z.hat[n]
```

Prediction: $\hat{Y}_{n+1} = Y_n + 0.43\hat{Z}_n = 8.076603$
--

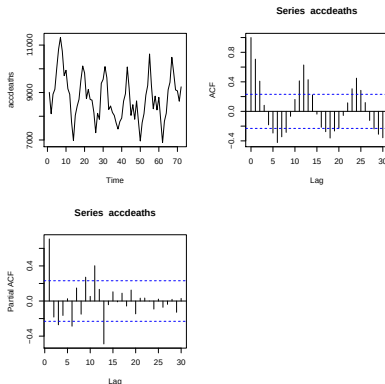
Example 2 - Accidental Death

This is an example concerning the number of accidental deaths on U.S. highways between 1973 and 1978. The data are stored in the file `accdeaths.dat`.

- Read data:

R-code:

```
par(mfrow=c(2,2))
accdeaths=scan("accdeaths.dat")
ts.plot(accdeaths)
acf(accdeaths,30)
pacf(accdeaths,30)
```



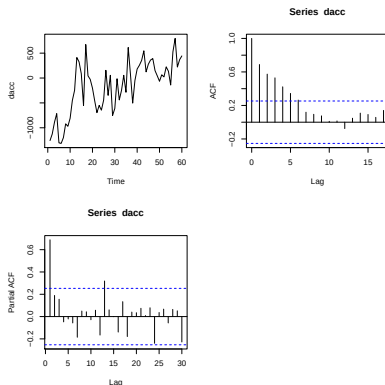
- The ACF graph shows a periodic component (seasonal pattern) at lag 12,24,36,...
- This motivates the use of seasonal differencing

Example 2 - Removing Trend and Seasonal Effect

- Seasonal differencing, $d = 12$

R-code:

```
dacc=diff(accdeaths,12)
par(mfrow=c(2,2))
ts.plot (dacc)
acf(dacc)
pacf(dacc,30)
```



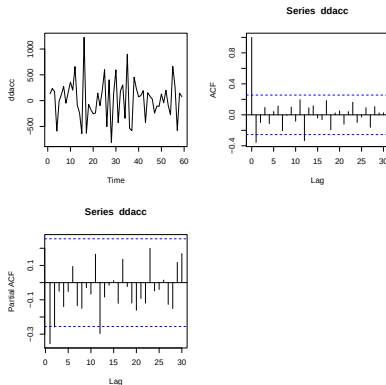
- The time series has a trend. Also, the ACF shows significant correlations, which motivates a further differencing

Example 2 - Removing Trend

- Further differencing

R-code:

```
ddacc=diff(dacc)
par(mfrow=c(2,2))
ts.plot(ddacc)
acf(ddacc,30)
pacf(ddacc,30)
```



- Significant values of ACF are detected at lags 1 and 12
 $\Rightarrow$ SARIMA(0,0,1) $\times$ (0,0,1)₁₂ model for the differenced data

Example 2 - Model fitting

Fitting $\text{SARIMA}(0, 0, 1) \times (0, 0, 1)_{12}$ model to the differenced data, $(1 - B)(1 - B^{12})X_t$

R-code:

```
ddacc.1=arima(ddacc - mean(ddacc), order=c(0,0,1),  
seasonal=list(order=c(0,0,1), period=12),include.mean=F)  
mean(ddacc)           # 28.8 #  
ddacc.1$coef[1]       # -0.48 #  
ddacc.1$coef[2]       # -0.59 #  
ddacc.1$aic           #854.9315#
```

Result :

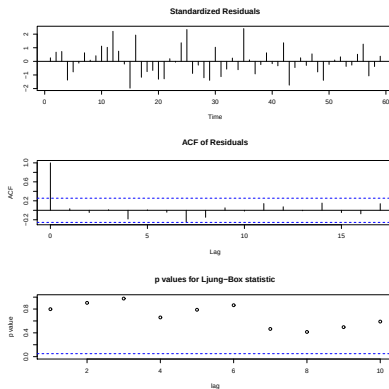
- $W_t = (1 - B)(1 - B^{12})X_t$
- $V_t = W_t - \overline{W}, \overline{W} = 28.8$
- $V_t = (1 - 0.49B)(1 - 0.59B^{12})Z_t$
- $\text{AIC} = 854.9315$

Example 2 - Diagnostic Checks

$$(1 - B)(1 - B^{12})X_t = 28.8 + (1 - 0.49B)(1 - 0.59B^{12})Z_t$$

R-Code:

```
tsdiag(ddacc.1)
```



Example 2 - Model Selection

Alternatively, instead of a $\text{SARIMA}(0,0,1) \times (0,0,1)_{12}$ model, we may fit an $\text{MA}(13)$ model

R-code:

```
ddacc.2<-arima(ddacc-mean(ddacc),order=c(0,0,13),  
include.mean=F,method="ML")  
ddacc.2$model$theta
```

$$\begin{aligned}\therefore (1 - B)(1 - B^{12})X_t \\ = 28.8 + (1 - 0.49B + 0.058B^2 \dots - 0.679B^{12} + 0.444B^{13})Z_t\end{aligned}$$

Example 2 - Model Selection

- There are 13 parameters in the MA(13) model

Q: Which of the parameters are significant?

A: Check if $\hat{\theta}_j$'s CI does not covers 0 $\Leftrightarrow$ Check if $|\frac{\hat{\theta}_j}{2\hat{\sigma}_j}| > 1$

R-code:

```
a=ddacc.2$model$theta/(2*sqrt(diag(ddacc.2$var.coef)))
```

a

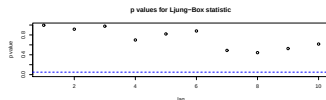
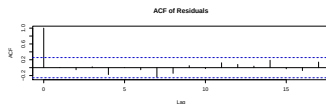
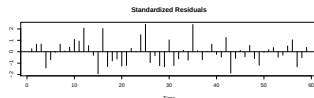
- Only $\theta_1, \theta_{12}, \theta_{13}$ and possibly θ_6 are significant

Example 2 - Model Selection

Refit the data by specifying $\theta_1, \theta_{12}, \theta_{13}$ to be the only nonzero values

R-code:

```
ddacc.3=arima(ddacc-mean(ddacc),  
include.mean=F, order=c(0,0,13),  
fixed=c(NA,0,0,0,0,0,0,0,0,0,0,NA,NA))  
ddacc.3$model$theta  
tsdiag(ddacc.3)  
ddacc.3$aic          # 856.6213#
```



$$\therefore (1 - B)(1 - B^{12})Z_t = 28.8 + (1 - 0.458B - 0.646B_{12} + 0.209B_{13})$$

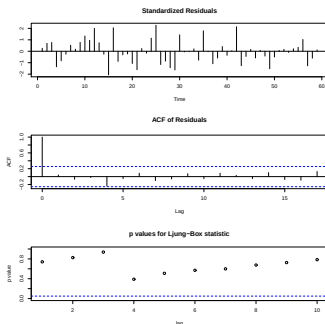
Example 2 - Model Selection

Fit also the model with $\theta_1, \theta_6, \theta_{12}, \theta_{13}$ nonzero

R-code:

```
ddacc.4=arima(ddacc-mean(ddacc),  
order=c(0,0,13), include.mean=F,  
fixed=c(NA,0,0,0,0,NA,0,0,0,0,0,NA,NA))  
ddacc.4$model$theta  
tsdiag(ddacc.4)  
ddacc.4$aic      # 854.4733 #
```

$$\begin{aligned} \therefore (1 - B)(1 - B^{12})X_t \\ = 28.8 + (1 - 0.596B - 0.406B^6 - 0.685B^{12} + 0.459B^{13})Z_t \end{aligned}$$



Example 2 - Model Selection

	ddacc.1	ddacc.2	ddacc.3	ddacc.4
Model	SARIMA(0, 0, 1) $\times$ (0, 0, 1) ₁₂	MA(13)	MA(13): $\theta_1, \theta_{12}, \theta_{13}$	MA(13): $\theta_1, \theta_6, \theta_{12}, \theta_{13}$
AIC	854.9315	867.91	856.62	854.4733

Looking at `tsdiag(ddacc.4)` :

- ACF plots shows no serial correlation
- **Ljung-Box test:** most of the test statistic has p-value over 0.05, which implies that the null hypothesis of the residual being WN is not inappropriate
- The final model is ddacc.4, i.e., SARIMA(0, 1, 13) $\times$ (0, 1, 0)₁₂
$$(1 - B)(1 - B^{12})X_t$$
$$= 28.8 + (1 - 0.596B - 0.406B^6 - 0.685B^{12} + 0.459B^{13})Z_t$$

```
library(forecast)
```

```
fit=Arima(accdeaths,order=c(0,1,13),seasonal = list(order = c(0, 1, 0),  
period = 12), fixed=c(NA,0,0,0,0,NA,0,0,0,0,0,NA,NA),include.constant=T)
```